

Introduction

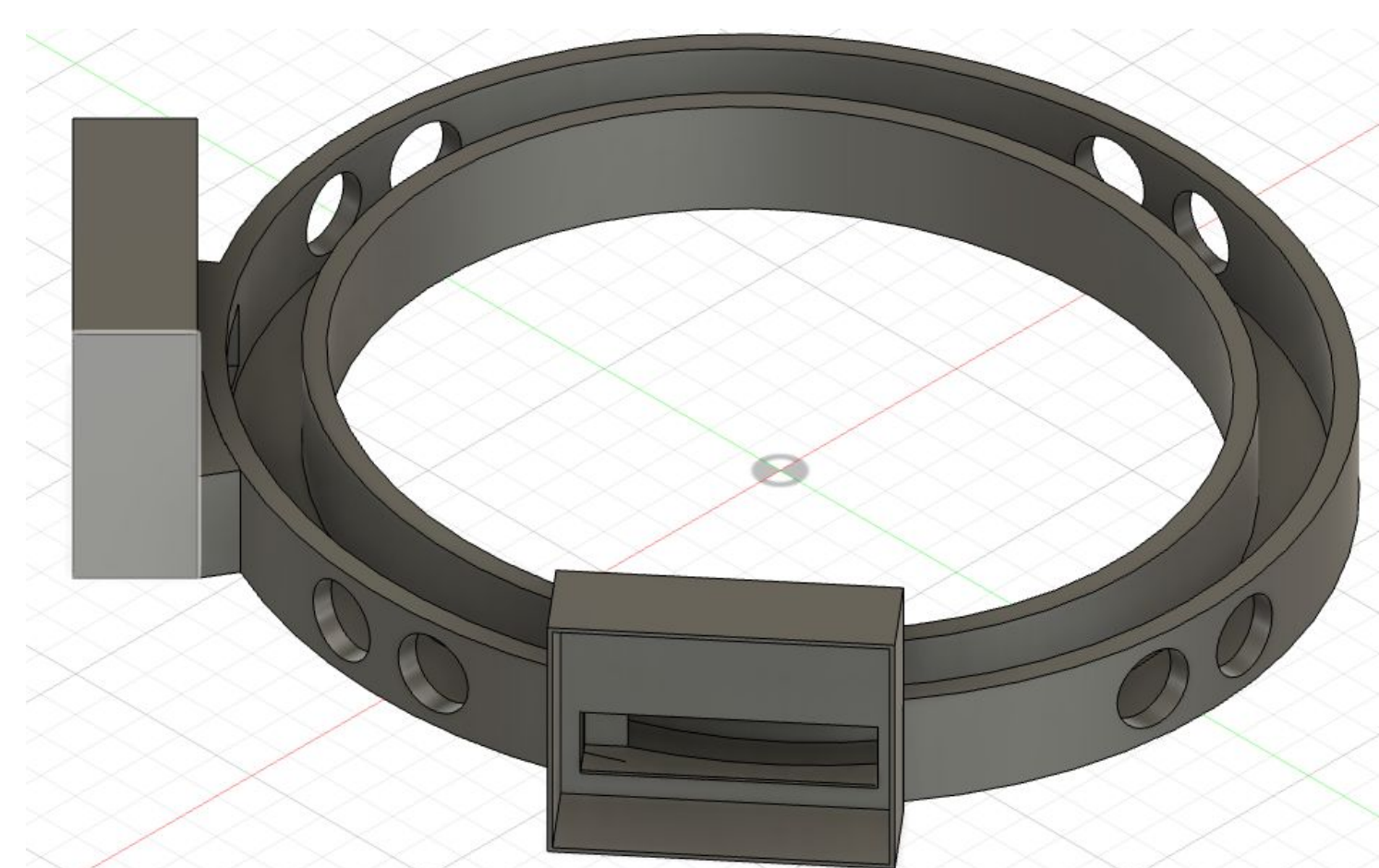
Head level obstacles are a problem for visually impaired people and there are not many affordable and accessible devices that help them with detecting these obstacles in complex environments.

Problem Statement

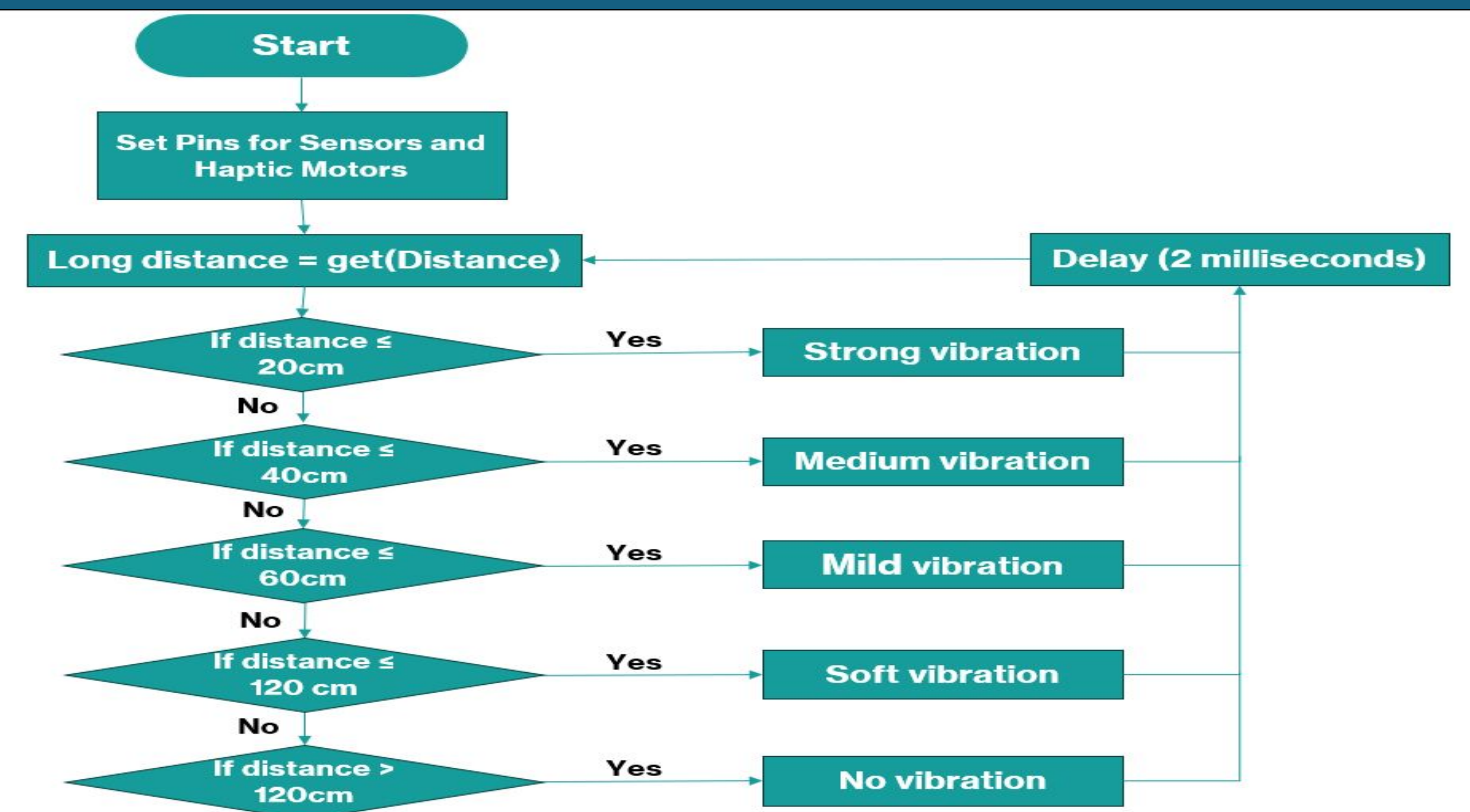
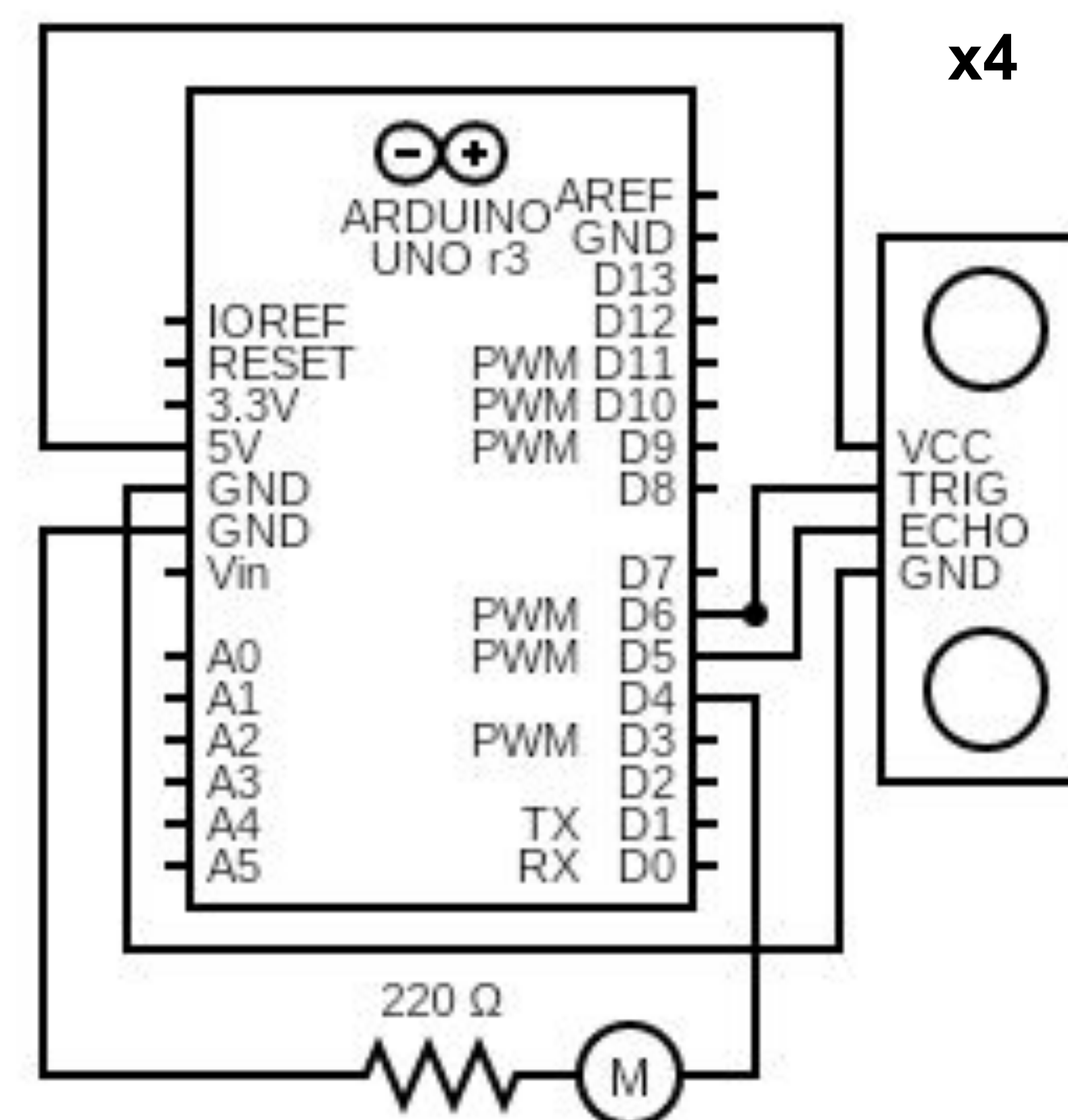
- Inadequate resources for visually impaired in impoverished areas few solutions allow visually impaired to detect/avoid obstacles at eye level that are available to them.
- Walking sticks/guide dogs aid awareness of obstacles that are in front of the person and on on the ground level
- Current solutions are often expensive/take excessive amounts of time to acquire and can be inaccessible for some communities
- Solution should be affordable, easy to use and effective.
- Design needs to help the visually impaired navigate obstacles at head level while being simple/effective using arduino electronics and basic materials.

Design Concept

- Use arduino and basic sensors to improve the quality of life of the visually impaired and improve the ability to navigate complex obstacles
- Making a device with the responsiveness and easy to use
- Accessible and affordable to all people in need



Circuit Diagram and Logic Flow Diagram



Test Results

- Tested prototype while standing stationary
- Tested prototype walking slowly, ~10 ft, with group members standing around the tester acting as obstacles
- Ultrasonic sensors were able to detect objects at the head level
- Haptic motors communicated with user and did increase vibration when object was close
- Worked best when object was within 40 cm of user
- Feedback from sensors was often inconsistent and delayed, especially when detecting objects moving away from it at distance greater than 40cm

Discussion of Test Results

- Prototype is wearable and functional. However size is somewhat small so not ideal for fitting all heads
- Some sensors were constantly vibrating, caused by possible sensory overload particularly in small or crowded environments
- Motor intensity was hard to distinguish and gauge a distance from

Conclusion and Future Work

- A more refined and comfortable housing unit to wear on the user's head is needed
- Both our code or sensors sensitivity should be thoroughly tested and adjusted in order to provide more reliable feedback through the motors for and ease of use
- Better sensing for wider coverage and not just cardinal directions
- An adjustable headwear design will be necessary in future iterations

References

- [1]"Arduino," Home. Accessed: Mar. 07, 2025. [Online]. Available: <https://www.arduino.cc/>
- [2]"Yours for the making," Instructables. Accessed: Mar. 07, 2025. [Online]. Available: <https://www.instructables.com/>

Acknowledgements

Thank you to Professor Vuong, Momchil Gavrilo, Payam Farhadi, and all ENG3 TA's for your assistance throughout the design process."